

# Usecase flow

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## Use-Case Flow Specification

Use Case: Ingest & Forward Webhook Payload (UC-01 & UC-03)

Primary Actor: External System

Secondary Actors: Target Service, System Operator

### *Preconditions:*

- The target endpoint URL and webhook subscription configurations exist and are active in the system.
- The Ingestion Hub service is online and ready to accept incoming HTTP requests.

### *Postconditions (Success):*

- The webhook payload is successfully delivered to the target service, and the transaction is logged with an HTTP 2xx response.

### *Postconditions (Failure):*

- The payload delivery fails after maximum retries, the payload is persisted in the Dead-Letter Queue (DLQ), and the incident is logged for operator review.

### *Main Success Scenario:*

- External System sends an inbound HTTP webhook event payload to the ingestion endpoint.
- System validates the payload signature using the configured secret (includes: UC-02 Validate Webhook Signature).
- System logs the incoming raw payload with a timestamp and assigns a unique event ID.
- System forwards the payload to the registered Target Service endpoint (UC-03 Forward Payload to Target).
- Target Service receives the payload and returns an HTTP 2xx success response.
- System updates the delivery log status to "SUCCESS" along with the response time metrics.
- Use case ends successfully.

*Alternate Flow 1 — 5a. Payload Delivery Fails (Extends to UC-04: Retry Failed Delivery)*

- 5a1. Target Service returns a non-2xx status code (e.g., 500/503) or the request times out.
- 5a2. System initiates exponential backoff retries at intervals of 1s, 2s, 4s, and 8s.
- 5a3. System re-attempts payload delivery to the Target Service up to a maximum of 4 total retry attempts.
- 5a4. If any retry attempt returns an HTTP 2xx response, the system logs the successful delivery and the flow resumes at step 6.
- 5a5. If all 4 retry attempts fail, the flow branches to Alternate Flow 2 (Extends to UC-05: Move to Dead-Letter Queue).

*Alternate Flow 2 — 5a5a. All Retries Exhausted (Extends to UC-05: Move to Dead-Letter Queue)*

- 5a5a1. System flags the webhook event as permanently failed.
- 5a5a2. System moves the payload and complete execution failure metadata into the Dead-Letter Queue (DLQ).
- 5a5a3. System logs the DLQ entry and alerts the System Operator for manual intervention.
- 5a5a4. Use case ends with failure state logged.